AXIOMS OF SET THEORY

„HOW $\in$ BEHAVES", „WHICH SETS EXIST"

ZERMELO-FRAENKEL THEORY (ZF) · 9 AXIOMS (7 AXIOMS + 2 SCHEMATA) (2 + 1 SCH. FOLLOW FROM THE OTHERS)

· OFTEN WE ADD 10. AXIOM - AXIOM OF CHOICE (AC) THE RESULTING THEORY „ZF + AC" = ZFC

1. $\boxed{\text{AXIOM OF EXISTENCE}}$ $(\exists x)(x = x)$ „A SET EXISTS".

2. $\boxed{\text{AXIOM OF EXTENSIONALITY}}$ $(\forall z)(z \in x \leftrightarrow z \in y) \rightarrow x = y$

· CONNECTION BETWEEN $=$, $\in$

„A SET IS DETERMINED BY ITS ELEMENTS"

„IF TWO SETS HAVE THE SAME ELEMENTS, THEY ARE EQUAL"

$\boxed{\text{EX.}}$ PROVE: $((x \subseteq y) \wedge (y \subset z)) \rightarrow x \subset z$

3. $\boxed{\text{AXIOM SCHEMA OF SEPARATION}}$ (COMPREHENSION): IF $\varphi(x)$ IS A FORMULA THAT DOES NOT CONTAIN z AS A FREE VARIABLE, THEN

(VYDĚLENÍ)

$(\forall a)(\exists z)(\forall x)(x \in z \leftrightarrow (x \in a \wedge \varphi(x)))$ IS AN AXIOM „FROM THE SET a WE SELECT ALL ELEMENTS WITH PROPERTY $\varphi(x)$, THEY WILL FORM A NEW SET z"

– BY THE AX. OF EXTENSIONALITY z IS UNIQUE

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NOTATION: $\{x;\ x\in a \wedge \varphi(x)\}$, SHORTLY $\{x\in a;\ \varphi(x)\}$ „THE SET OF ALL ELEMENTS OF a SATISFYING $\varphi(x)$"

DEF INTERSECTION $a\cap b$ IS $\{x,\ x\in a \wedge x\in b\}$ [$\cap$, $\setminus$ ARE BINARY FUNCTIONAL SYMBOLS OPERATORS]

SET DIFFERENCE $a\setminus b$ $(a-b)$ IS $\{x,\ x\in a \wedge x\notin b\}$

THE EMPTY SET $\emptyset$ IS $\{x,\ x\in a \wedge x\neq x\}$ WHERE a IS AN ARBITRARY SET

(IT IS UNIQUE BY AX. OF EXTENSIONALITY)

EX · WRITE USING A FORMULA „A SET a HAS EXACTLY ONE ELEMENT."

4. **AXIOM OF A PAIR** (PAIRING) $(\forall a)(\forall b)(\exists z)(\forall x)(x\in z \longleftrightarrow (x=a \vee x=b))$

„FOR EVERY SETS a,b THERE EXISTS A SET z WHOSE ELEMENTS ARE EXACTLY a AND b". — z IS UNIQUE

NOTATION: $\{a,b\}$ – UNORDERED PAIR OF SETS a,b. IF $a\neq b$ .. TWO-ELEMENT SET CONTAINING a,b

$\{a\}$ MEANS $\{a,a\}$ .. ONE-ELEMENT SET CONTAINING a.

EXAMPLE · WE CAN CREATE $\{\emptyset\}, \{\{\emptyset\}\}, \{\{\{\emptyset\}\}\}, \{\emptyset,\{\emptyset\}\}, \ldots$

EX PROVE

$$(\forall z)(x\in z \Leftrightarrow y\in z) \rightarrow x=y$$

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DEF. (a,b) – <u>ORDERED PAIR</u> OF SETS a, b IS THE SET $\{\{a\},\{a,b\}\}$ $\quad \{a,\{a,b\}\}$

$\langle a,b \rangle$

<u>REMARK</u>. IF $a=b$ THEN $(a,a)=\{\{a\},\{a,a\}\}=\{\{a\},\{a\}\}=\{\{a\}\}$

LEMMA $(x,y)=(m,n) \iff (x=m \wedge y=n)$

<u>PROOF</u>. ($\Leftarrow$). IF $x=m$ THEN $\{x\}=\{m\}$ BY THE AX. OF EXTENSIONALITY

$\{x,y\}=\{m,n\}$ – // –

SO $\{\{x\},\{x,y\}\}=\{\{m\},\{m,n\}\}$ – // –

($\Rightarrow$) $\{\{x\},\{x,y\}\}=\{\{m\},\{m,n\}\}$ THEN $\{x\}=\{m\}$ OR $\{x\}=\{m,n\}$. IN BOTH CASES $x=m$

(BY SUBSTITUTION)

- $\{m,n\}=\{x\}$ OR $\{m,n\}=\{x,y\}$ SO $n=x$ OR $n=y$.
- IF $n=x$ THEN FROM $x=m$ WE HAVE $\underline{n}=m=x=\underline{y}$
- IF $\underline{n=y}$ $\square$

DEF IF $a_1, a_2, \dots, a_n$ ARE SETS, WE DEFINE THE ORDERED n-TUPLE $(a_1, a_2, \dots, a_n)$ $\langle a_1, a_2, \dots, a_n \rangle$ AS FOLLOWS:

(a_1) MEANS a_1.

IF $(a_1, \dots, a_k)$ IS DEFINED, THEN $(a_1, a_2, \dots, a_k, a_{k+1})$ IS $((a_1, \dots, a_k), a_{k+1})$

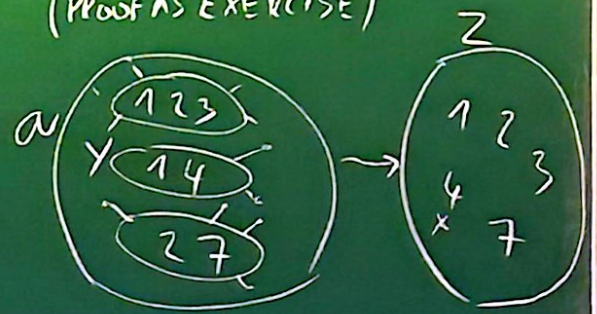
LEMMA $(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n) \Longleftrightarrow (a_1 = b_1 \wedge a_2 = b_2 \wedge \dots \wedge a_n = b_n)$ (PROOF AS EXERCISE)

5. **AXIOM OF UNION** $(\forall a)(\exists z)(\forall x)(x \in z \Leftrightarrow (\exists y)(x \in y \wedge y \in a))$

z = SUMY

DEF. $\bigcup a$ IS "$\{x, (\exists y)(x \in y \wedge y \in a)\}$" THE UNION OF THE SET a (SUMA)

UNARY OPERATOR



OBSERVATION IF $a = \{b, c\}$ THEN $\bigcup a = \bigcup \{b, c\} =$ "$\{x, x \in b \vee x \in c\}$"

DEF. $b \cup c$ IS $\bigcup \{b, c\}$. THE UNION OF SETS b AND c (SJEDINJENJE) $\cup$ .. A BINARY OPERATOR

DEF. IF $a_1, a_2, \dots, a_n$ ARE SETS, WE DEFINE THE UNORDERED n-TUPLE $\{a_1, a_2, \dots, a_n\}$ (n-ELEMENT SET IF ALL ARE a_i ARE DISTINCT) RECURSIVELY:

IF $\{a_1, \dots, a_k\}$ IS DEFINED, THEN $\{a_1, a_2, \dots, a_k, a_{k+1}\}$ IS $\{a_1, a_2, \dots, a_k\} \cup \{a_{k+1}\}$

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